

UNIT-2

→ The 5G wireless propagation channels :

- Channel model requirements
- Propagation scenarios & challenges in the 5G modeling
- Channel Models for mmWave
- MIMO Systems

OFDM / VFMC / FBMC ¹ ~~CP~~ ² ~~FL~~

Feature	OFDM	VFMC	FBMC
Complexity	Low	medium	high
Latency	Low	medium	high
Fattening	Low	medium	high
Implementation Difficulty	Low	medium	high
Robustness (strong)	Low	medium	high
Flexibility	Low	medium	high
energy efficiency	Low	medium	high
Interference	High	Medium	Low
Frequency	high	medium	Low
Scalability	Medium	High	High
Signal Quality	Medium	High	High
Sustainability	4G	5G	5G
Type	Modulation	Modulation	Modulation
Orthogonality	Yes	Yes	Yes
Basic Idea	Divide data into sub carriers	Filter sub-bands	Filter each sub carrier
used in 4G	Yes	NO	NO
used in 5G	Yes	Yes	Yes

OFDM vs VFMC vs FBMC

Feature	OFDMA	NOMA
Complexity	Low	High
Latency	Low	Low
Fading	Low	Low
Implementation Difficulty	Low	High
Robustness (Strong)	Medium	High
Flexibility	Medium	High
energy efficiency	Low	High
Interference	Low	High
Frequency	Medium	High
Scalability	Medium	High
Signal Quality	Medium	High
Sustainability	4G	5G
Type	Multiple Access	Multiple Access
Orthogonality	Yes	No
Basic idea	Divide Subcarriers	Share Power Domain
Used in 4G	Yes	No
Used in 5G	Yes	Yes

* Multi-Carrier Design in 5G

(Design Principles for Multi-user Commⁿ)

⇒ 5G requires:

- High data rate
- Low latency
- Massive connectivity

→ To achieve this, multi-carrier design is used.

⇒ In this technique:

- Data is divided into multiple parallel subcarriers (frequencies)
- Each subcarrier carries a part of data.

⇒ Multi-carrier design is a transmission technique

- where data is split into multiple streams & transmitted simultaneously
- using multiple orthogonal subcarriers.

→ Support different services:

- eMBB (high speed)
- URLLC (Low latency)
- mMTC (IoT devices)

Design Principles

①. Orthogonality

- Subcarriers should not interfere with each other

②. High Spectral efficiency
- use bandwidth efficiently

③. Flexibility

- support multiple services with diff requirements

ex: IoT $\rightarrow$ low data

Video Streaming $\rightarrow$ high data

④. Low Latency

⑤. Robustness $\rightarrow$ Should handle Doppler shift, noise, Fading

⑥. Scalability

⑦. Energy efficiency $\rightarrow$ Power consumption should be low

⑧. Implementation $\rightarrow$ should be easy

⑨. Multi-users Support

$\Rightarrow$ It is a technique

- where data is transmitted using multiple Subcarriers

$\rightarrow$ Used in:

- OFDM

- FBMC

- UFMC

* Spectral Efficiency in 5G

Intro

- Spectrum is limited
- Users are increasing
- ⇒ So, 5G focuses on high Spectral efficiency
- ⇒ Spectral Efficiency is the amount of data transmitted per unit bandwidth.

Units: bits/sec/Hz

Need

- Limited Spectrum
- high demand
- Need to support many users

Techniques to Improve Spectral Efficiency

1. Advanced waveforms

- OFDM, FBMC, UFMC
- Better utilization of Spectrum

2. NOMA

- Multiple users share same frequency
- increases Capacity

3. Massive MIMO

- Multiple antennas

4. Beamforming

- signal directed towards user
- improves signal strength

5. Resource Allocation

- Dynamic allocation of bandwidth

6. Small cells

- Divide large area into small cells.

Adv

- High data rate
- massive connected
- improve performance

dis

- Complex
- high cost
- interference

Spectrum

- ~~Spectrum~~ Space for signal transmission
- Space for signal to move

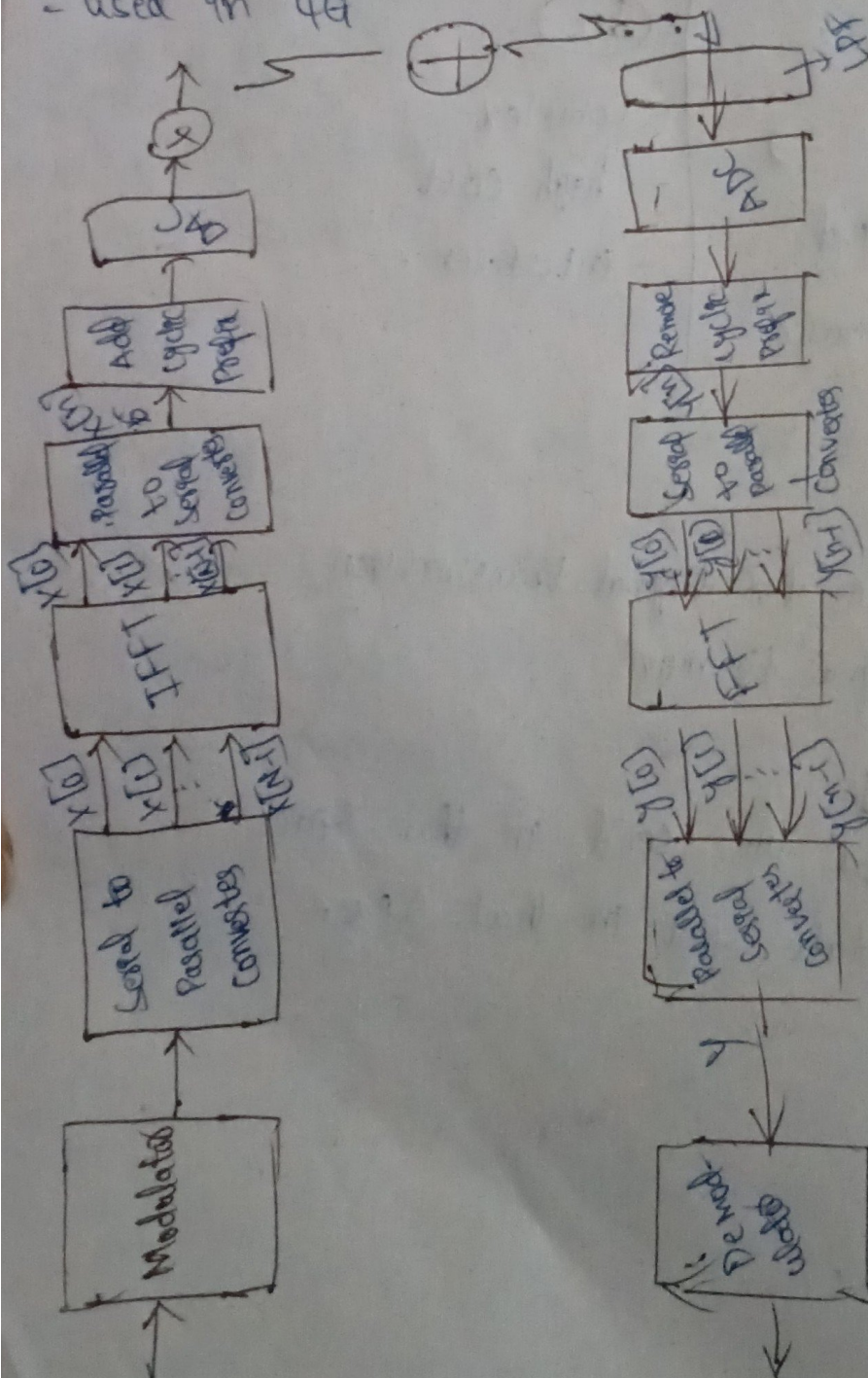
Spectrum efficiency

- how much data we send in that space
- amount of data sent in that space

① OFDM - Orthogonal Frequency Division Multiplexing

→ It is a technique

- ~~that~~ divides data into many small signals & send them on different frequencies (subcarriers)
- Uses multiple subcarriers
- Subcarriers are orthogonal (no interference)
- Uses Cycle Prefix
- used in 4G



②. UFMC - Universal Filtered Multi-carrier

→ improved version of OFDM.

→ Groups of subcarriers are filtered

— to reduce noise.

— Less noise than OFDM

— No full cycle prefix